

DEVELOPMENT OF A COHERENT DIFFRACTIVE IMAGING ENDSTATION AT THE SPring-8 II 12XU

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Abstract

The National Synchrotron Radiation Research Center (NSRRC) operates two beamlines, 12B2 and 12XU, at the SPring-8 synchrotron in Japan. With the scheduled for upgrade to SPring-8 II, significant improvements in beam coherence, stability, and brilliance are anticipated. While the Taiwan Photon Source (TPS) and SPring-8 II are both advanced facilities, SPring-8 II provides distinct advantages for high-energy and coherence-demanding applications, offering complementary capabilities to TPS. To harness these improvements, NSRRC is constructing a dedicated Coherent Diffractive Imaging (CDI) endstation at beamline 12XU. Building on successful CDI demonstrations at PETRA III and earlier feasibility studies at SPring-8, the new endstation is located downstream of the inelastic X-ray scattering endstation. Adjustable slits are installed upstream to optimize beam size and intensity, while Kirkpatrick-Baez mirrors are retracted during CDI experiments to preserve coherence. The endstation includes two main subsystems: a zone plate-based X-ray microscope and a detector assembly. The zone plate-based X-ray microscope, mounted on a granite base, integrates modular optics and a nanometer-resolution piezo-driven sample stage with granite or Invar components for thermal stability. A precision linear guideway enables flexible detector positioning over a 1.5 m range, ensuring high-resolution imaging and experimental versatility.

INTRODUCTION

Synchrotron radiation has become a fundamental tool for exploring matter at the nanoscale, driven by advances in fourth-generation light sources that deliver higher beam coherence, stability, and brightness. Among emerging imaging methodologies, Coherent Diffractive Imaging (CDI) stands out for its high-resolution capabilities and reconstructing both amplitude and phase information from diffraction patterns via iterative algorithms. CDI has shown remarkable application in many fields, J. Miao's pioneering work extended crystallography to non-crystalline specimens [1]. More recently, a comprehensive review in Nature showcased how CDI and ptychography have unified microscopy and crystallography, enabling imaging from atomic to centi-meter scales, including 3D atomic defect structures and ultrafast dynamics in quantum and energy materials [2]. CDI's applicability extends across scientific disciplines. In materials science, Bragg CDI (BCDI) has

been used to image three-dimensional strain distribution within individual nanocrystals with sub-nanometer accuracy and study how strain affects device performance [3]. Furthermore, CDI has been successfully applied to investigate strain and structural faults in single nanowires [4]. These examples underline CDI's ability to capture fine structural and dynamic details easily. Inspired by these methodological advances and grounded in the feasibility demonstrated at reference facilities, NSRRC is now constructing a CDI endstation on beamline 12XU at SPring-8. This marks a strategic expansion of Taiwan's synchrotron research, leveraging cutting-edge beam quality and instrumentation to advance next-generation imaging science.

CURRENT STATUS OF TESTING CDI ENDSTATION AT SPring-8 12XU

At the current stage of testing, the CDI endstation is installed downstream of the SPring-8 12XU inelastic X-ray scattering (IXS) endstation, positioned directly behind the IXS detector. The source is 4.5 m long in-vacuum undulator with a magnet period of 32 mm. To accommodate the present parameters of the zone plate, the zone plate-based microscope is mounted on a linear guide rail that allows free translation along the beam axis, thereby enabling fine adjustment of the endstation to its optimal position. The linear rail has a length exceeding 2.5 m, and both the CDI endstation and the downstream detector are supported on granite bases (450 mm x 450 mm x 1065 mm). Once translated to the desired location, the bases are stabilized by lowering Hi-wedge® HWM C-1111 leveling feet, which fill the gap between the granite and the experimental floor and are then securely locked in place to maintain system stability. Since the Hi-wedges only serve to bridge the gap, additional XYZ adjustment stages are mounted on top of the granite bases for fine manual alignment of the endstation and detector modules. These stages provide a travel range of ± 8 mm in the horizontal direction and ± 10 mm in the vertical direction. During CDI experiments, the upstream IXS endstation, together with the analyzer and the Kirkpatrick-Baez (K-B) mirrors, can be retracted from the X-ray beam path when required, as shown in Fig. 1. This configuration ensures that the CDI setup receives a direct and coherent beam while preserving the operational flexibility of the upstream instrumentation. Under standard operating conditions, the distance between the detector and the sample is approximately 1 m.

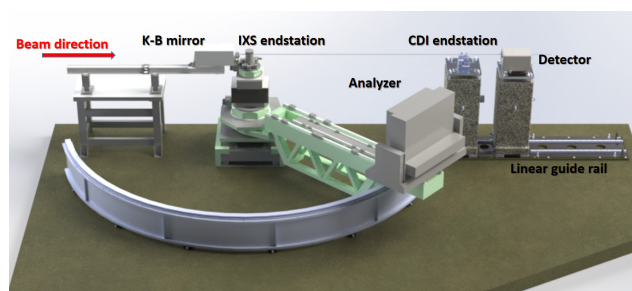


Figure 1: The downstream beamline optics of SPring-8 12XU.

The zone plate-based microscope comprises several modular components, including: (a) a beam stopper, (b) a zone plate, (c) an Order-Sorting Aperture (OSA), and (d) a sample positioning stage, see Fig. 2. Each module is equipped with an independent XYZ piezo-driven translation stage, offering a travel range of ± 6 mm and positioning resolution down to the nanometer scale. The modules are mainly assembled from SmarAct[®] XYZ-SLC 17:22, SmarAct[®] XY-SLC 5252, and SmarAct[®] XYZ-CHS 5237. The additional flexure stage, PI[®] P-616.3C, is adopted for faster sample scanning. To further mitigate thermal drift and ensure dimensional stability, the stage bases can be constructed from Invar alloy, which offers low thermal expansion characteristics critical to maintaining optical alignment over extended experimental durations.

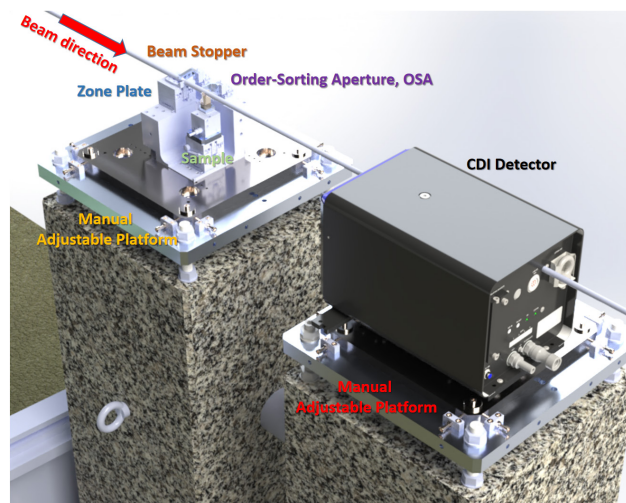


Figure 2: Setup of CDI endstation and detector at SPring-8 12XU.

BACKUP CDI ENDSTATION AT NSRRC TPS 25

Since the SPring-8 II upgrade is still underway, a test CDI endstation has been constructed at the TPS 25A beamline to evaluate the present performance of CDI under TPS conditions and to facilitate convenient testing. This backup setup enables us to assess and predict the expected performance of the CDI endstation following the SPring-8 upgrade.

The source is 3.0 m long in-vacuum undulator with a magnet period of 22 mm at TPS 25A. The CDI endstation is located in the middle-to-downstream section of the

beamline. After passing through the double-crystal monochromator (DCM), the X-ray beam successively traverses the slits, the first horizontal focusing mirror (HFM1), the horizontal secondary source (HSS) slit, the vertical focusing mirror (VFM), and the horizontal focusing mirror (HFM). The partially focused beam then enters the CDI endstation.

Because the backup CDI endstation must share the same X-ray path with the existing endstation, a laterally translatable linear rail was designed, incorporating precision positioning blocks. This arrangement ensures that when switching between different endstations, the optical components do not require extensive realignment, thereby minimizing setup time. Once the granite base is translated into position, it is securely fixed to the underlying platform to maintain overall system stability.

To reduce the spacing between the adjustment stages of individual optical components, SmarAct[®] XYZ-SLC 17:22 stages were adopted for the zone plate, the order-sorting aperture (OSA), and the beam stopper. In order to investigate the effect of varying zone plate focal lengths, the sample stage requires a larger travel range along the optical axis. For this purpose, PI[®] N-765 (XY) and PI[®] N-565 (Z) stages were implemented. In addition, for specialized experiments, a compact angular rotation platform was integrated within the 8 mm clearance beneath the flexure stage used for sample scanning. The rotation axis employs a weak-link structure combined with a Piezo LEGS[®] 20N actuator, providing an angular range of approximately ± 10 mrad with a resolution of 10 nrad. As shown in Fig. 3. In general, the detector is positioned approximately 1.5 m downstream of the sample. For clarity, the detector is not shown in the figure below.

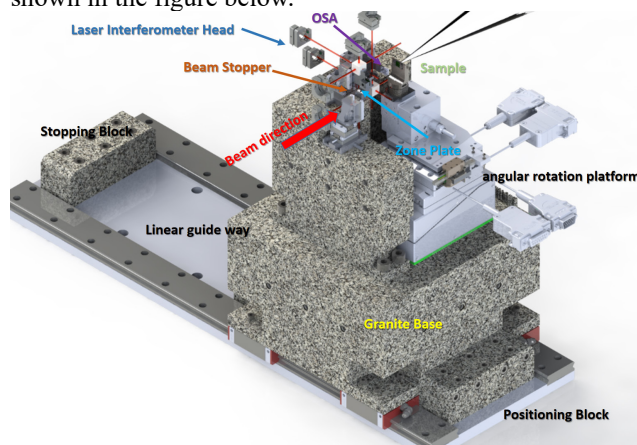


Figure 3: Setup of backup CDI endstation at NSRRC TPS 25.

RESULTS AND DISCUSSIONS

A Siemens star test pattern was used as the standard resolution sample for evaluating the imaging performance of both the CDI endstations at SPring-8 and NSRRC. To ensure a consistent basis for comparing imaging resolution, the incident X-ray energy was fixed at 8.98 keV during the tests. As shown in the Fig. 4, the spatial resolution of the CDI endstations reached 4.4 nm at SPring-8 12XU and

10.0 nm at NSRRC TPS 25A, respectively. These results confirm the feasibility of the current reconstruction methodology and experimental setup.

The superior resolution achieved at SPring-8 12XU may be attributed to several factors: (1) the higher degree of spatial coherence at the SPring-8 light source, which improves the signal-to-noise ratio; (2) the higher photon flux available at the experimental energy range, enabling shorter acquisition times; (3) the choice of zone plate optics and sample stage, which influences their relative mechanical stability; (4) the stability of the experimental floor and the extent of air turbulence in the experimental hutch; and (5) differences in the control of ambient temperature, which affect thermal drift and system alignment. (6) the overall stability of optical mirrors along the beam path, which has a direct impact on the final image quality.

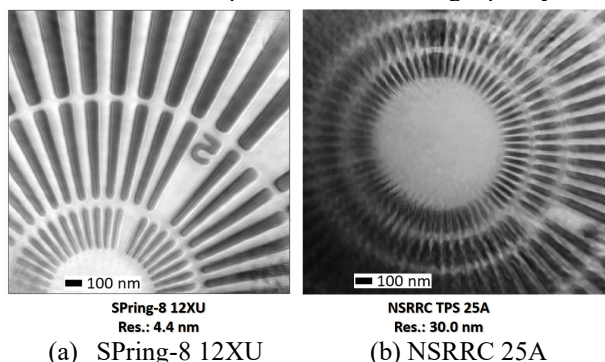


Figure 4: Siemens star test pattern and spatial resolution of the two CDI endstations.

CONCLUSION

Based on the measurement results, both CDI endstations were shown to effectively achieve nanometer-scale imaging resolution. This outcome provides strong confidence in the construction of the CDI endstation following the SPring-8 upgrade.

The upgraded SPring-8 is expected to offer higher beam coherence, superior floor stability compared to TPS, and reduced optical interference along the beam path, all of these reasons will create better conditions for CDI experiments. Furthermore, the significant performance improvements in the high-energy range will complement the TPS endstation, enabling a broader spectrum of scientific investigations.

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